Author Manuscript
Author Manuscript
Author Manuscript
Author Manuscript

HHS Public Access
Author manuscript
Pediatr Pulmonol. Author manuscript; available in PMC 2022 September 09.

Published in final edited form as:
Pediatr Pulmonol. 2019 June ; 54(6): 680–682. doi:10.1002/ppul.24309.

Severe Childhood Asthma Exacerbations: Is Treatment Response Variability in the Genes?

Monica Tang[1], **Kanecia Zimmerman**[1,2], **Jason E. Lang**[1,2]

[1]Department of Pediatrics, Duke University School of Medicine, Durham, North Carolina.

[2]Duke Clinical Research Institute, Duke University School of Medicine, Durham, North Carolina.

Abstract

Worldwide, asthma is one of the most common causes of medical emergency in children. Failed acute management events comprise a large part of the annual healthcare costs of asthma. These severe exacerbations requiring hospitalization likely also contribute to permanent remodeling and impaired lung function in later life. Various studies have uncovered clinical, environmental, and genetic risk factors for severe asthma and exacerbations that cannot be acutely managed and require hospitalization. Tse and colleagues extend their previous work by adding important new insights into the determinants of failed ED management of pediatric asthma. Using a candidate gene approach and stepwise regression, they identified three SNPs in two candidate genes: *IL33*, rs7037276, rs1342326 and *SPATS2L*, rs295137, which when aggregated together significantly increased the odds for ED management failure, with each risk allele increasing failure odds by 83% (95% CI, 36–145%). When these genes markers were combined with validated clinical predictors of acute asthma management failure (viral infection, poor baseline pediatric respiratory assessment measure score, oxygen saturation <92%, fever >38.3°C, and presence of symptoms between episodes) the resulting ability to predict acute management failure was significantly improved (ROC curve=0.82). As discussed in this editorial and the discussion by Tse et al, these novel genetic markers provide new avenues of research for management of acute exacerbations. As the syndrome of asthma continues to be better characterized into multiple endo-phenotypes, we are likely to find that personalized treatment is necessary not only for *control* of asthma but also for acute *rescue* of exacerbations.

Editorial

Around the world, asthma is one of the most common causes of medical emergency and hospitalization in children. The *International Asthma Insights and Reality Survey*[1] found that in the United States (US), up to 9% of patients with asthma require hospitalization each year. Similar admission rates were 7%, 15% and 20% in Western Europe, Asia and Central/Eastern Europe, respectively. In large, multiethnic populations, asthma hospitalization disproportionately affects poor children and children from racial and ethnic minorities. African-American children suffer hospitalization and death rates roughly 4 and 10 times higher than White US children[2]. Of the 7–8 million US children with asthma, about half experience ≥1 exacerbations each year resulting, in about 1 million emergency department (ED) visits and 200,000 hospitalizations annually - just due to asthma[3,4]. Best practice for acute severe exacerbations includes escalating use of short-acting beta-agonists and

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

systemic corticosteroids. In the ED setting, most exacerbations can be reversed; however, a significant portion fail ED management and require hospitalization and even intensive care. These failed ED management events comprise a large part of the annual healthcare costs of asthma, which total $50 billion in the United States and $2 billion in Canada[5]. These events also contribute to permanent respiratory damage and airway remodeling[6], which are now recognized collectively as a likely contributor to chronic obstructive pulmonary disease (COPD) later in life[6,7]. Due to these substantial economic costs and health sequelae, identification of risk factors predicting ED management failure and hospitalization is an area of intense research.

Numerous studies have identified clinical, environmental, and genetic factors that influence the risk for severe asthma exacerbations and hospitalization[8–10]. In 1981, Fischl and colleagues studied adolescents and young adults presenting to the ED with acute asthma to determine predictors of further emergency care and hospitalization[11]. Of the 205 patients, 45 (22%) required hospitalization and 40 (20%) had relapse. The investigators created a multifactorial predictor 'Fischl' index – taking into account tachycardia, tachypnea, pulses paradoxus, peak expiratory flow rate, dyspnea, accessory muscle use, and wheezing - which successfully predicted relapse and admission in about 95% of subjects. Analysis from the Childhood Asthma Management Program (CAMP) study of children 5–12 years of age with mild to moderate asthma, found that risk factors for asthma-related hospitalization included early asthma onset, longer duration of asthma, need for asthma controller therapy, airflow obstruction, and lower patient IQ[12].

More recently, in a cohort study of 278 children spanning 1–17 years of age seen in the ED with acute asthma, Keogh and colleagues found previous intensive care admission, initial oxygen saturation, asthma symptom score, and albuterol dosing more often than every hour were most helpful in predicting hospitalization[13]. With ≥3 predictors present, there was a 92% likelihood for hospitalization. Collectively, past studies demonstrate that the degree of respiratory dysfunction at presentation (exacerbation severity) and prior history of severe exacerbations portend high risk for future acute management failure and hospitalization.

A notable environmental predictor of pediatric asthma hospitalization includes detection of a respiratory viral pathogen, which has been associated with 80% of asthma exacerbations in children. Additionally, fall season, indoor secondhand smoke exposure, and poor ambient air-quality also increase the risk for hospitalization[14–17]. More recently, genome wide association studies have identified important genetic loci. Bonnelykke and colleagues linked data from children with recurrent acute hospitalizations for asthma between 2–6 years of age in the Danish National Patient Registry and DNA from the Danish Neonatal Screening Biobank to identify several variants associated with severe exacerbation risk. Loci included *IL33*, interleukin receptor like 1 (*IL1RL1*), and cadherin-related family member 3 (*CDHR3*)[18]. Genetic influences have been shown to associate with severe exacerbations in the setting of certain exposures (i.e. gene x environment interaction), including mold (chitinase 1 gene *CHIT1*)[19], dust mite (transforming growth factor-beta 1 gene *TGFB1*)[20], picornavirus infection (interferon regulator factor 7 *IRF7*)[21] and chronic beta-agonist drug exposure (β2-adrenoreceptor gene *ADRB2*[22]).

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

In this issue of *Pediatric Pulmonology*, Tse et al.[23] extend their recent work by adding important insights into the determinants of failed ED management of pediatric asthma. They aimed to combine both clinical and genetic information to better predict ED management failure, defined as hospitalization, ED stay ≥ 8 hours after oral corticosteroid intake, or a return ED visit for asthma within 72 hours. Subjects were selected from the previously published *Determinants Of Oral corticosteroid Responsiveness in Wheezing Asthmatic Youth* (DOORWAY) study[9,24], which determined the best clinical predictors of ED management failure in 1–17 year old children presenting with acute asthma[9]. These predictors were viral detection, baseline pediatric respiratory assessment measure (PRAM) score, oxygen saturation <92%, fever >38.3°C, and presence of symptoms between episodes.

In the current study, Tse and colleagues restricted their analysis to Caucasian children with complete clinical and genetic data to address the hypothesis that a key factor in determining response to acute asthma treatment stems from human genetic variation. Using multivariable logistic regression, they interrogated 53 SNPs, chosen as candidates due to their previous association with asthma development, asthma phenotypes, or response to asthma medication. Five SNPs demonstrating individual association with ED management failure with a p-value <0.1 were included in multivariable stepwise regression with previously-identified clinical determinants and study site.

Of the 491 subjects studied, most were male and below 6 years of age and 89 (18%) failed ED management. Backward stepwise regression identified resulted in three SNPs in two genes: *IL33*, rs7037276, rs1342326 and *SPATS2L*, rs295137. With the combination of the clinical predictors plus these new genetic loci data, the predictive value was significantly (albeit modestly) improved to an area under the ROC curve=0.82, which is considered "good" accuracy for a diagnostic test[25]. Aggregated together, these risk alleles significantly increased the odds for ED management failure, with each risk allele increasing failure odds by 83% (95% CI, 36–145%).

Interpretation of these findings must consider several points. Since exacerbation risk is highest in young children, this study included many children <6 years old where diagnosis of asthma is often challenging and is likely to include some mixture of mechanistically-distinct wheezing phenotypes. Analysis was restricted to the Caucasian children, and requires replication in other ancestral populations, particularly those at high-risk for asthma morbidity. The importance of the findings of Tse and colleagues is not in its bedside ability to predict management response, but rather its novel genetic insights into the response heterogeneity among children treated for acute exacerbation.

Studies of this kind are extremely difficult to complete, and the author deserve commendation; however future studies are needed to replicate these genetic findings, preferably with more homogenous phenotypes (i.e. atopic asthma or obesity-related asthma) and with additional racial groups.

What (if anything) are *IL33* and *SPATS2L* telling us about pediatric management failure? *IL33* resides on chromosome 9p24.1 and encodes the eponymous cytokine of the IL-1 family which activates the Th2 pathway by binding to the interleukin 1 receptor-like 1

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

(IL1RL1)[26,27]. In asthma, expression of IL33 is upregulated in bronchial epithelium and airway smooth muscle cells[28,29], while SNPs in the IL33/IL1RL1 pathway (including rs7037276) associate with wheeze presenting during toddlerhood[30]. *SPATS2L* resides on chromosome 2q33.1 and is expressed in many tissues, including the lung and may protect against oxidative stress in myoblasts via ribosome biogenesis and translational control[27,31]. A recent GWAS analyzing bronchodilator responsiveness in non-Hispanic white asthmatics found the strongest association within *SPATS2L* at the same rs295137 found here by Tse[32]. Knockdown of mRNA led to increased β2 adrenergic receptor (β2R) protein levels, suggesting that *SPATS2L* is a key β2R regulator and likely determinant of exacerbation management responsiveness. As the authors rightly note, these findings will need replication and validation by functional studies. As the syndrome of asthma continues to be better characterized into multiple endo-phenotypes, we are likely to find that personalized treatment is necessary not only for *control* of asthma but also for acute *rescue* of exacerbations.

References:

1. Rabe KF, Adachi M, Lai CK, Soriano JB, Vermeire PA, Weiss KB, Weiss ST. Worldwide severity and control of asthma in children and adults: the global asthma insights and reality surveys. J Allergy Clin Immunol 2004;114(1):40–47. [PubMed: 15241342]
2. Akinbami LJ, Moorman JE, Garbe PL, Sondik EJ. Status of childhood asthma in the United States, 1980–2007. Pediatrics 2009;123 Suppl 3:S131–145. [PubMed: 19221156]
3. Moorman JE, Akinbami LJ, Bailey CM, Zahran HS, King ME, Johnson CA, Liu X. National surveillance of asthma: United States, 2001–2010. Vital & health statistics Series 3, Analytical and epidemiological studies 2012(35):1–58.
4. CDC - Asthma - Data and Surveillance - Asthma Surveillance Data. 2018.
5. Nurmagambetov T, Kuwahara R, Garbe P. The Economic Burden of Asthma in the United States, 2008–2013. Ann Am Thorac Soc 2018;15(3):348–356. [PubMed: 29323930]
6. Matsunaga K, Hirano T, Oka A, Tanaka A, Kanai K, Kikuchi T, Hayata A, Akamatsu H, Akamatsu K, Koh Y, Nakanishi M, Minakata Y, Yamamoto N. Progression of Irreversible Airflow Limitation in Asthma: Correlation with Severe Exacerbations. J Allergy Clin Immunol Pract 2015;3(5):759–764 e751. [PubMed: 26054551]
7. Martinez FD. Early-Life Origins of Chronic Obstructive Pulmonary Disease. N Engl J Med 2016;375(9):871–878. [PubMed: 27579637]
8. Dougherty R, Fahy JV. Acute Exacerbations of Asthma: Epidemiology, Biology and the Exacerbation-Prone Phenotype. Clin Exp Allergy 2009;39(2):193–202. [PubMed: 19187331]
9. Ducharme FM, Zemek R, Chauhan BF, Gravel J, Chalut D, Poonai N, Guertin MC, Quach C, Blondeau L, Laberge S, network DrgotPERiC. Factors associated with failure of emergency department management in children with acute moderate or severe asthma: a prospective, multicentre, cohort study. Lancet Respir Med 2016;4(12):990–998. [PubMed: 27451347]
10. Forno E, Fuhlbrigge A, Soto-Quiros ME, Avila L, Raby BA, Brehm J, Sylvia JM, Weiss ST, Celedon JC. Risk factors and predictive clinical scores for asthma exacerbations in childhood. Chest 2010;138(5):1156–1165. [PubMed: 20472862]
11. Fischl MA, Pitchenik A, Gardner LB. An index predicting relapse and need for hospitalization in patients with acute bronchial asthma. N Engl J Med 1981;305(14):783–789. [PubMed: 7266631]
12. Wu AC, Tantisira K, Li L, Schuemann B, Weiss ST, Fuhlbrigge AL. Predictors of symptoms are different from predictors of severe exacerbations from asthma in children. Chest 2011;140(1):100–107. [PubMed: 21292760]
13. Keogh KA, Macarthur C, Parkin PC, Stephens D, Arseneault R, Tennis O, Bacal L, Schuh S. Predictors of hospitalization in children with acute asthma. J Pediatr 2001;139(2):273–277. [PubMed: 11487756]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

14. Etzel RA. How environmental exposures influence the development and exacerbation of asthma. Pediatrics 2003;112(1 Pt 2):233–239. [PubMed: 12837915]
15. Johnston SL, Pattemore PK, Sanderson G, Smith S, Lampe F, Josephs L, Symington P, O'Toole S, Myint SH, Tyrrell DA, et al. Community study of role of viral infections in exacerbations of asthma in 9–11 year old children. BMJ (Clinical research ed) 1995;310(6989):1225–1229.
16. McCarville M, Sohn MW, Oh E, Weiss K, Gupta R. Environmental tobacco smoke and asthma exacerbations and severity: the difference between measured and reported exposure. Archives of disease in childhood 2013;98(7):510–514. [PubMed: 23606711]
17. Teach SJ, Gergen PJ, Szefler SJ, Mitchell HE, Calatroni A, Wildfire J, Bloomberg GR, Kercsmar CM, Liu AH, Makhija MM, Matsui E, Morgan W, O'Connor G, Busse WW. Seasonal risk factors for asthma exacerbations among inner-city children. J Allergy Clin Immunol 2015;135(6):1465–1473.e1465. [PubMed: 25794658]
18. Bonnelykke K, Sleiman P, Nielsen K, Kreiner-Moller E, Mercader JM, Belgrave D, den Dekker HT, Husby A, Sevelsted A, Faura-Tellez G, Mortensen LJ, Paternoster L, Flaaten R, Molgaard A, Smart DE, Thomsen PF, Rasmussen MA, Bonas-Guarch S, Holst C, Nohr EA, Yadav R, March ME, Blicher T, Lackie PM, Jaddoe VW, Simpson A, Holloway JW, Duijts L, Custovic A, Davies DE, Torrents D, Gupta R, Hollegaard MV, Hougaard DM, Hakonarson H, Bisgaard H. A genome-wide association study identifies CDHR3 as a susceptibility locus for early childhood asthma with severe exacerbations. Nature genetics 2014;46(1):51–55. [PubMed: 24241537]
19. Wu AC, Lasky-Su J, Rogers CA, Klanderman BJ, Litonjua AA. Fungal exposure modulates the effect of polymorphisms of chitinases on emergency department visits and hospitalizations. Am J Respir Crit Care Med 2010;182(7):884–889. [PubMed: 20538957]
20. Sharma S, Raby BA, Hunninghake GM, Soto-Quiros M, Avila L, Murphy AJ, Lasky-Su J, Klanderman BJ, Sylvia JS, Weiss ST, Celedon JC. Variants in TGFB1, dust mite exposure, and disease severity in children with asthma. Am J Respir Crit Care Med 2009;179(5):356–362. [PubMed: 19096005]
21. Bosco A, Ehteshami S, Panyala S, Martinez FD. Interferon regulatory factor 7 is a major hub connecting interferon-mediated responses in virus-induced asthma exacerbations in vivo. J Allergy Clin Immunol 2012;129(1):88–94. [PubMed: 22112518]
22. Turner S, Francis B, Vijverberg S, Pino-Yanes M, Maitland-van der Zee AH, Basu K, Bignell L, Mukhopadhyay S, Tavendale R, Palmer C, Hawcutt D, Pirmohamed M, Burchard EG, Lipworth B. Childhood asthma exacerbations and the Arg16 beta2-receptor polymorphism: A meta-analysis stratified by treatment. J Allergy Clin Immunol 2016;138(1):107–113.e105. [PubMed: 26774659]
23. Tse SM, Krajinovic M, Chauhan BF, Zemek R, Gravel J, Chalut D, Poonai N, Quach C, Laberge S, Ducharme FM, network DrgotPERiC. Genetic determinants of acute asthma therapy response in children with moderate-to-severe asthma exacerbations. Pediatr Pulmonol 2019.
24. Ducharme FM, Zemek R, Gravel J, Chalut D, Poonai N, Laberge S, Quach C, Krajinovic M, Guimont C, Lemiere C, Guertin MC. Determinants Of Oral corticosteroid Responsiveness in Wheezing Asthmatic Youth (DOORWAY): protocol for a prospective multicentre cohort study of children with acute moderate-to-severe asthma exacerbations. BMJ Open 2014;4(4):e004699.
25. Tape TG. Interpreting Diagnostic Tests. University of Nebraska Medical Center2019.
26. Schmitz J, Owyang A, Oldham E, Song Y, Murphy E, McClanahan TK, Zurawski G, Moshrefi M, Qin J, Li X, Gorman DM, Bazan JF, Kastelein RA. IL-33, an interleukin-1-like cytokine that signals via the IL-1 receptor-related protein ST2 and induces T helper type 2-associated cytokines. Immunity 2005;23(5):479–490. [PubMed: 16286016]
27. Online Mendelian Inheritance in Man (OMIM). 2019.
28. Prefontaine D, Nadigel J, Chouiali F, Audusseau S, Semlali A, Chakir J, Martin JG, Hamid Q. Increased IL-33 expression by epithelial cells in bronchial asthma. J Allergy Clin Immunol. Volume 125. United States2010. p 752–754. [PubMed: 20153038]
29. Yagami A, Orihara K, Morita H, Futamura K, Hashimoto N, Matsumoto K, Saito H, Matsuda A. IL-33 mediates inflammatory responses in human lung tissue cells. Journal of immunology (Baltimore, Md : 1950) 2010;185(10):5743–5750.
30. Savenije OE, Mahachie John JM, Granell R, Kerkhof M, Dijk FN, de Jongste JC, Smit HA, Brunekreef B, Postma DS, Van Steen K, Henderson J, Koppelman GH. Association of IL33-

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

IL-1 receptor-like 1 (IL1RL1) pathway polymorphisms with wheezing phenotypes and asthma in childhood. The Journal of allergy and clinical immunology 2014;134(1):170–177. [PubMed: 24568840]

31. Zhu CH, Kim J, Shay JW, Wright WE. SGNP: an essential Stress Granule/Nucleolar Protein potentially involved in 5.8s rRNA processing/transport. PLoS One 2008;3(11):e3716. [PubMed: 19005571]
32. Himes BE, Jiang X, Hu R, Wu AC, Lasky-Su JA, Klanderman BJ, Ziniti J, Senter-Sylvia J, Lima JJ, Irvin CG, Peters SP, Meyers DA, Bleecker ER, Kubo M, Tamari M, Nakamura Y, Szefler SJ, Lemanske RF, Zeiger RS, Strunk RC, Martinez FD, Hanrahan JP, Koppelman GH, Postma DS, Nieuwenhuis MAE, Vonk JM, Panettieri RA, Markezich A, Israel E, Carey VJ, Tantisira KG, Litonjua AA, Lu Q, Weiss ST. Genome-Wide Association Analysis in Asthma Subjects Identifies SPATS2L as a Novel Bronchodilator Response Gene. PLoS Genet. Volume 82012.